

Reporte de QA E2E local

Stafford WhatsApp y Easterns Automotive Group - evidencia reproducible

Resultado	Alcance	Evidencia
PASS	Stafford WhatsApp: 7/7; frontend local PASS	JSON de evidencia + captura
PASS	Easterns: 8/8 rutas solicitadas	JSON + SQL de verificacion
PASS	Regresion: 328 tests + 4 stress	Suite local completa
BLOCKED	Ninguno	Sin dependencias pendientes

Resumen ejecutivo

Se ejecuto un E2E completamente local usando el PostgreSQL de Docker proporcionado, una API local y el frontend local. Los Customer Replied fueron simulados y firmados con un HMAC de prueba inyectado por variables de entorno; el secreto no se imprimio ni se guardo en esta evidencia.

Stafford WhatsApp quedo conforme con la regla especial: el telefono ya existe, pero el backend solo crea la cola cuando interpreta un vehiculo. Sin vehiculo permanece parcial. Se probaron 15 segundos de estabilizacion, 30 minutos dentro de horario y 3 horas fuera de horario con reloj simulado. Easterns quedo dirigido por datos de locations, sin listas de ciudades codificadas en el servicio.

Action Pre Owned Cars queda verificado con los dos nombres solicitados: Action Pre Owned Cars English y Action Pre Owned Cars Español.

Version exacta probada y entorno

Elemento	Valor
Commit publicado	15d20a2c8b6b2646264a28f63a745321677349a6
Rama / remoto	main / https://github.com/juanfer93/dealeradmin.git
BD	dealeradmin-postgres; c20d4b05ae50; postgres:18; Up; puerto 5432
Migracion ultima	EasternsLocationRoutingRules1710000021000
API / frontend	http://127.0.0.1:3010 / http://127.0.0.1:3000
Zona horaria	America/Bogota
Config no secreta	NODE_ENV=development; API_URL local; reloj de prueba; namespace determinista
Externos	GHL=false; Neon=false; cuentas reales=false
Generado	2026-09-12T17:55:26+00:00

La tabla locations tiene 5,298 registros: outside_md_va 4,074; virginia 688; maryland_laurel 316; southern_md_overlap 218; baltimore_overlap 1; silver_spring_laurel 1.

Matriz AC-01 a AC-20

ID	Criterio	Estado	Evidencia / timestamp
AC-01	DB, API y frontend locales saludables	PASS	API 3010 + frontend 3000; 2026-09-12
AC-02	Migraciones verificadas en PostgreSQL	PASS	Ultima 1710000021000; SELECT migrations
AC-03	Webhook Stafford WhatsApp firmado	PASS	7 casos; HTTP 201; IDs deterministas
AC-04	Telefono preexistente no basta solo	PASS	phone-without-vehicle -> partial; sin lead_dealer
AC-05	Vehiculo interpretado habilita minima	PASS	Mustang y truck en snapshot
AC-06	Contaminantes no inventan categoria	PASS	requisitos/financiar/Mustang aislado
AC-07	Incompleta en horario: +30 min	PASS	10:00 Bogota -> 15:30Z; queued
AC-08	Incompleta fuera: +3 h	PASS	20:00 Bogota -> 04:00Z; queued
AC-09	Poll de estabilizacion +15 s	PASS	process-due con X-Test-Now

ID	Criterio	Estado	Evidencia / timestamp
AC-10	Nombre y vehiculo separados	PASS	real_name y vehicle_type independientes
AC-11	Webhook duplicado idempotente	PASS	duplicate_ignored; un mensaje
AC-12	Firma invalida rechazada	PASS	HTTP 401
AC-13	Baltimore rota Rosedale/Laurel	PASS	Caso 1 Rosedale; caso 2 Laurel
AC-14	New Jersey -> Rosedale	PASS	Newark NJ
AC-15	Silver Spring -> Laurel	PASS	Catalogo MD
AC-16	Washington/DC -> Sterling	PASS	Estado explicito DC
AC-17	Southern Maryland rota	PASS	Waldorf Laurel; La Plata Sterling
AC-18	Virginia -> Sterling	PASS	Richmond VA
AC-19	Ruteo consulta localidades BD	PASS	5,298 locations; zona + indice
AC-20	Regresion + Action bilingue	PASS	328/328 + 4/4 stress; nombres OK

Detalle de cada conversacion Stafford WhatsApp

Cada caso usa conversation_id y event_id deterministas dentro del namespace qa-whatsapp-stafford-20260912. Los webhooks se enviaron al endpoint local y la evidencia correlaciona conversation_messages, webhook_events, conversations, leads y lead_dealers mediante SQL.

Caso	Mensajes	Estado/snapshot	Ventana
day-vehicle-only	¿Qué requisitos necesito? busco un Mustang	none -> queued; vehicle=Mustang; phone=+12405552001	10:00; +15 s; +30 min -> queued
night-vehicle-only	busco un truck	none -> queued; vehicle=truck; phone=+12405552002	20:00; +15 s; +3 h -> queued
phone-without-vehicle	Hola, quiero información	none -> partial; vehicle=-; phone=+12405552003	partial
declared-name-and-vehicle	Me llamo Sofía Stafford quiero una Honda Civic	none -> waiting_window; vehicle=Honda Civic; phone=+12405552004	waiting_window
contaminated-replies	¿Qué requisitos necesito? Quiero financiar un auto Mustang	none -> partial; vehicle=-; phone=+12405552005	partial
duplicate-webhook	busco un SUV busco un SUV	none -> waiting_window; vehicle=SUV; phone=+12405552006	waiting_window

qa-whatsapp-stafford-20260912-day-vehicle-only

event_id=qa-whatsapp-stafford-20260912-day-vehicle-only-event-01 | message_ids=8c3faa36-93ee-4bb8-869d-b717e9bf72cc | payload_hash=b2797e35104020d33543d27
event_id=qa-whatsapp-stafford-20260912-day-vehicle-only-event-02 | message_ids=8c3faa36-93ee-4bb8-869d-b717e9bf72cc,5e2cf716-fdb2-4c5b-a2e4-3c4c3db24aa0 |

Estado: queued; snapshot={"assigned_dealer_id": "3bf48365-6833-4fb7-b401-1d6ea04ba975", "bank_account": "", "documents": "", "down_payment": "", "identification": "", "message_count": 2, "missing_qualification": ["real_name", "down_payment", "purchase_timeline", "identification", "proof_of_income", "bank_account"], "phone": "+12405552001", "purchase_timeline": "", "qualification_complete": false, "qualification_memory": "vehicle: Mustang", "real_name": "", "vehicle_type": "Mustang"}; location_snapshot={"city": null, "easterns_zone": null, "state": null, "zip_code": null}; next_attempt_at=None

Ventana simulada release_at=2026-09-12T15:30:00.000Z; transicion confirmada por poll.

SQL de verificacion:

```
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.channel, c.status,
       c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, c.ready_at,
       l.id AS lead_id, l.canonical_phone, l.first_name, l.last_name,
       ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = $1
ORDER BY ld.created_at NULLS LAST
```

qa-whatsapp-stafford-20260912-night-vehicle-only

event_id=qa-whatsapp-stafford-20260912-night-vehicle-only-event-01 | message_ids=f6c5c755-a090-416c-8002-f104ce96bb57 | payload_hash=c2c1587cc174766ba653a

Estado: queued; snapshot={"assigned_dealer_id": "3bf48365-6833-4fb7-b401-1d6ea04ba975", "bank_account": "", "documents": "", "down_payment": "", "identification": "", "message_count": 1, "missing_qualification": ["real_name", "down_payment", "purchase_timeline", "identification", "proof_of_income", "bank_account"], "phone": "+12405552002", "purchase_timeline": "", "qualification_complete": false, "qualification_memory": "vehicle: truck", "real_name": "", "vehicle_type": "truck"}; location_snapshot={"city": null, "easterns_zone": null, "state": null, "zip_code": null}; next_attempt_at=None

Ventana simulada release_at=2026-09-13T04:00:00.000Z; transicion confirmada por poll.

SQL de verificacion:

```
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.channel, c.status,
       c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, c.ready_at,
       l.id AS lead_id, l.canonical_phone, l.first_name, l.last_name,
       ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = $1
ORDER BY ld.created_at NULLS LAST
```

qa-whatsapp-stafford-20260912-phone-without-vehicle

event_id=qa-whatsapp-stafford-20260912-phone-without-vehicle-event-01 | message_ids=13da86f0-e8cc-471b-85b8-3441681755ff | payload_hash=3e20ba8d85aa5483a

Estado: partial; snapshot={"assigned_dealer_id": "3bf48365-6833-4fb7-b401-1d6ea04ba975", "bank_account": "", "documents": "", "down_payment": "", "identification": "", "message_count": 1, "missing_qualification": ["real_name", "vehicle_type", "down_payment", "purchase_timeline", "identification", "proof_of_income", "bank_account"], "phone": "+12405552003", "purchase_timeline": "", "qualification_complete": false, "qualification_memory": "", "real_name": "", "vehicle_type": ""}; location_snapshot={"city": null, "easterns_zone": null, "state": null, "zip_code": null}; next_attempt_at=None

SQL de verificacion:

```
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.channel, c.status,
       c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, c.ready_at,
       l.id AS lead_id, l.canonical_phone, l.first_name, l.last_name,
       ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
```

```
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = $1
ORDER BY ld.created_at NULLS LAST
```

qa-whatsapp-stafford-20260912-declared-name-and-vehicle

```
event_id=qa-whatsapp-stafford-20260912-declared-name-and-vehicle-event-01 | message_ids=d84e1c5d-f401-40b9-82fb-e20cf161b5b8 | payload_hash=9a35a06ad68675
event_id=qa-whatsapp-stafford-20260912-declared-name-and-vehicle-event-02 | message_ids=d84e1c5d-f401-40b9-82fb-e20cf161b5b8, a2bde05c-aa5f-4c46-bab0-264f0
```

Estado: waiting_window; snapshot={"assigned_dealer_id": "3bf48365-6833-4fb7-b401-1d6ea04ba975", "bank_account": "", "documents": "", "down_payment": "", "identification": "", "message_count": 2, "missing_qualification": ["down_payment", "purchase_timeline", "identification", "proof_of_income", "bank_account"], "phone": "+12405552004", "purchase_timeline": "", "qualification_complete": false, "qualification_memory": "real_name: Sofia Stafford; vehicle: Honda Civic", "real_name": "Sofia Stafford", "vehicle_type": "Honda Civic"}; location_snapshot={"city": null, "easterns_zone": null, "state": null, "zip_code": null}; next_attempt_at=2026-09-12T15:00:15.000Z

SQL de verificación:

```
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.channel, c.status,
       c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, c.ready_at,
       l.id AS lead_id, l.canonical_phone, l.first_name, l.last_name,
       ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = $1
ORDER BY ld.created_at NULLS LAST
```

qa-whatsapp-stafford-20260912-contaminated-replies

```
event_id=qa-whatsapp-stafford-20260912-contaminated-replies-event-01 | message_ids=24219e7f-4471-41aa-bf53-7272e3567384 | payload_hash=e8386f8f60f95a824a5
event_id=qa-whatsapp-stafford-20260912-contaminated-replies-event-02 | message_ids=24219e7f-4471-41aa-bf53-7272e3567384, c7570fff-484e-4d66-abbd-73a0db9dd
event_id=qa-whatsapp-stafford-20260912-contaminated-replies-event-03 | message_ids=24219e7f-4471-41aa-bf53-7272e3567384, c7570fff-484e-4d66-abbd-73a0db9dd
```

Estado: partial; snapshot={"assigned_dealer_id": "3bf48365-6833-4fb7-b401-1d6ea04ba975", "bank_account": "", "documents": "", "down_payment": "", "identification": "", "message_count": 3, "missing_qualification": ["real_name", "vehicle_type", "down_payment", "purchase_timeline", "identification", "proof_of_income", "bank_account"], "phone": "+12405552005", "purchase_timeline": "", "qualification_complete": false, "qualification_memory": "real_name": "", "vehicle_type": ""}; location_snapshot={"city": null, "easterns_zone": null, "state": null, "zip_code": null}; next_attempt_at=None

SQL de verificación:

```
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.channel, c.status,
       c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, c.ready_at,
       l.id AS lead_id, l.canonical_phone, l.first_name, l.last_name,
       ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = $1
ORDER BY ld.created_at NULLS LAST
```

qa-whatsapp-stafford-20260912-duplicate-webhook

```
event_id=qa-whatsapp-stafford-20260912-duplicate-webhook-event-01 | message_ids=ce1852d0-b38d-4000-837c-8b12c7e6aba1 | payload_hash=c631864469c8b85bfc72cc
event_id=qa-whatsapp-stafford-20260912-duplicate-webhook-event-01 | message_ids=ce1852d0-b38d-4000-837c-8b12c7e6aba1 | payload_hash=c631864469c8b85bfc72cc
```

Estado: waiting_window; snapshot={"assigned_dealer_id": "3bf48365-6833-4fb7-b401-1d6ea04ba975", "bank_account": "", "documents": "", "down_payment": "", "identification": "", "message_count": 1, "missing_qualification": ["real_name", "down_payment", "purchase_timeline", "identification", "proof_of_income", "bank_account"], "phone": "+12405552006", "purchase_timeline": "", "qualification_complete": false, "qualification_memory": "vehicle: SUV", "real_name": "", "vehicle_type": "SUV"}; location_snapshot={"city": null, "easterns_zone": null, "state": null, "zip_code": null}; next_attempt_at=2026-09-12T15:00:15.000Z

SQL de verificación:

```
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.channel, c.status,
       c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, c.ready_at,
       l.id AS lead_id, l.canonical_phone, l.first_name, l.last_name,
       ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = $1
ORDER BY ld.created_at NULLS LAST
```

Detalle de cada conversacion Easterns

La secuencia usa ocho conversaciones Messenger independientes en easterns. locations aporta state_code y easterns_routing_zone; la rotacion consulta el historial persistido de lead_dealers.

Caso	Ciudad/estado	Dealer	Routing reason	Estado
baltimore-1	Baltimore / MD	Rosedale	Baltimore Overlap: Round-Robin (Previous: Laurel/None)	PASS
baltimore-2	Baltimore / MD	Laurel	Baltimore Overlap: Round-Robin (Previous: Rosedale)	PASS
new-jersey	Newark / NJ	Rosedale	Exclusive Zone: State NJ	PASS
silver-spring	Silver Spring / MD	Laurel	Exclusive Zone: Maryland catalog → Laurel	PASS
washington	Washington / DC	Sterling	Exclusive Zone: State District of Columbia	PASS
southern-md-1	Waldorf / MD	Laurel	Southern MD/DC Overlap: Round-Robin (Previous: Sterling/None)	PASS
southern-md-2	La Plata / MD	Sterling	Southern MD/DC Overlap: Round-Robin (Previous: Laurel)	PASS
virginia	Richmond / VA	Sterling	Exclusive Zone: State Virginia	PASS

qa-easterns-routing-20260912-baltimore-1

Mensajes: Me llamo Buyer Uno | busco un Mustang | tengo 1000 de enganche | compro esta semana | Estoy en Baltimore, MD

```
conversation_id=qa-easterns-routing-20260912-baltimore-1-conversation; event_ids=qa-easterns-routing-20260912-baltimore-1-event-01,
qa-easterns-routing-20260912-baltimore-1-event-02, qa-easterns-routing-20260912-baltimore-1-event-03, qa-easterns-routing-20260912-baltimore-1-event-04,
qa-easterns-routing-20260912-baltimore-1-event-05; message_ids=319cce03-0639-45e4-b38e-b2b5c106e74f, 2d9d129e-0724-4a62-9894-f198ea8d7f1e,
5c524fce-56a3-4be4-a96e-a6e3a3156ada, 3ea566be-e5c1-4308-931d-e249a9ba5ac5, e15c17e1-6477-4388-9834-7972e236f117

payload_hashes=36d6ba5a0f148a3e2bfb0e67c42176bc0f9d134bbf3e541240c8819af854a351, 6648264756376de00ef2be26dff745a104854ccc6b28fc08abcc6fe3b5984a5d,
2ef6d5a8e313e949393ebdbb31bf14addeb072c372a3d4828729b55ae0f2e3fb, 30d05af743872311466f0e83451b5680b8d424ffccbae8af67a5e6345f4dcfb25,
a795c33ee76a4303dab537a06b176fc31f1d13793434675a28e2856cf7deb0f0

snapshot={"assigned_dealer_id": "d1111111-1111-1111-1111-111111111111", "bank_account": "", "documents": "", "down_payment": "1000",
"identification": "", "message_count": 5, "missing_qualification": ["identification", "proof_of_income", "bank_account"], "phone": "+12405553100",
"purchase_timeline": "this week", "qualification_complete": true, "qualification_memory": "real_name: Buyer Uno; vehicle: Mustang; down payment: 1000;
timeline: this week", "real_name": "Buyer Uno", "vehicle_type": "Mustang"; location_snapshot={"city": "Baltimore", "easterns_zone": "baltimore", "state":
"MD", "zip_code": null}; dealer=Rosedale; next_attempt_at=None
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.status,
c.qualification_snapshot, c.location_snapshot, c.next_attempt_at,
l.id AS lead_id, l.canonical_phone, ld.dealer_id, ld.assigned_dealer_id,
ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = $1
ORDER BY ld.created_at NULLS LAST
```

qa-easterns-routing-20260912-baltimore-2

Mensajes: Me llamo Buyer Dos | busco un Mustang | tengo 1000 de enganche | compro esta semana | Estoy en Baltimore, MD

```
conversation_id=qa-easterns-routing-20260912-baltimore-2-conversation; event_ids=qa-easterns-routing-20260912-baltimore-2-event-01,
qa-easterns-routing-20260912-baltimore-2-event-02, qa-easterns-routing-20260912-baltimore-2-event-03, qa-easterns-routing-20260912-baltimore-2-event-04,
qa-easterns-routing-20260912-baltimore-2-event-05; message_ids=4310512f-9ee2-483b-8240-3fab613073db, a7dfe0a7-d0f8-44f7-b8a0-f850827d5f81,
7dc7eec5-dffa-4dab-9e86-5e2e4d65b722, e35956b0-8f4f-4cc9-a7d5-b755c760ca41, 166ace6a-4883-4107-9b98-1b737fe52a54

payload_hashes=2359fc953699a989cdd10dd22d8bd409124540081398674d597d29c6c3f72110, 3e712d5a9952c19699fb717d5e5592bcb6f3fe2c99b4b3f791d4ecc752f77f1f3,
b09a055fd1cebca596ee9ee2be677f8380cc0d126a5f6e26178081600d284ec, 9ab9cc5a2c851dfcaa10546295d84916eb7ce4420705e8e3bf167072a224fef3,
a0fd8fb42d6c376cf7d85edb02fa0487e6cac1f92cd18805b01b41577a051ae2

snapshot={"assigned_dealer_id": "d1111111-1111-1111-1111-111111111111", "bank_account": "", "documents": "", "down_payment": "1000",
"identification": "", "message_count": 5, "missing_qualification": ["identification", "proof_of_income", "bank_account"], "phone": "+12405553101",
"purchase_timeline": "this week", "qualification_complete": true, "qualification_memory": "real_name: Buyer Dos; vehicle: Mustang; down payment: 1000;
timeline: this week", "real_name": "Buyer Dos", "vehicle_type": "Mustang"; location_snapshot={"city": "Baltimore", "easterns_zone": "baltimore", "state":
"MD", "zip_code": null}; dealer=Laurel; next_attempt_at=None
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.status,
c.qualification_snapshot, c.location_snapshot, c.next_attempt_at,
l.id AS lead_id, l.canonical_phone, ld.dealer_id, ld.assigned_dealer_id,
ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = $1
ORDER BY ld.created_at NULLS LAST
```

qa-easterns-routing-20260912-new-jersey

Mensajes: Me llamo Buyer Tres | busco un Mustang | tengo 1000 de enganche | compro esta semana | Estoy en Newark, NJ

```
conversation_id=qa-easterns-routing-20260912-new-jersey-conversation; event_ids=qa-easterns-routing-20260912-new-jersey-event-01,
qa-easterns-routing-20260912-new-jersey-event-02, qa-easterns-routing-20260912-new-jersey-event-03, qa-easterns-routing-20260912-new-jersey-event-04,
qa-easterns-routing-20260912-new-jersey-event-05; message_ids=054c0b6a-67ec-4903-97c7-4a507c98ed19, 7d0d74b5-3c46-46d0-b7d4-84ea8bf56925,
```

e132ea23-ddef-443a-9380-fbf8b8e5ebd5, 65ddc652-fd45-4a57-a662-ab2882801e99, 62975b58-3136-4ebb-942d-2d8877f52290

payload_hashes=29818a0bd32b9a726a90831a0f2fa039dd3959f6771c3f5bf339e0d2137f539a, b0b6b9e51874af5eee94f35f849d3ebbef612c1a9e42788880a5942bc37910de, 35ddfa85645c388d0fcea52ce6bf45c3d2add3467919cddf07ade2be24f698b, 0d89c69af58e26393f52835d280793d381c2a088f3cc0f88d419e66996c9db91, 74b4ad734f4bb97298172ed1e2df77b52b6a457eec020253f42a50e5359b741

snapshot={"assigned_dealer_id": "d1111111-1111-1111-1111-111111111111", "bank_account": "", "documents": "", "down_payment": "1000", "identification": "", "message_count": 5, "missing_qualification": ["identification", "proof_of_income", "bank_account"], "phone": "+12405553102", "purchase_timeline": "this week", "qualification_complete": true, "qualification_memory": "real_name: Buyer Tres; vehicle: Mustang; down payment: 1000; timeline: this week", "real_name": "Buyer Tres", "vehicle_type": "Mustang"}; location_snapshot={"city": "Newark", "easterns_zone": null, "state": "NJ", "zip_code": null}; dealer=Rosedale; next_attempt_at=None
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.status, c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, l.id AS lead_id, l.canonical_phone, ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = \$1
ORDER BY ld.created_at NULLS LAST

qa-easterns-routing-20260912-silver-spring

Mensajes: Me llamo Buyer Cuatro | busco un Mustang | tengo 1000 de enganche | compro esta semana | Estoy en Silver Spring, MD

conversation_id=qa-easterns-routing-20260912-silver-spring-conversation; event_ids=qa-easterns-routing-20260912-silver-spring-event-01, qa-easterns-routing-20260912-silver-spring-event-02, qa-easterns-routing-20260912-silver-spring-event-03, qa-easterns-routing-20260912-silver-spring-event-04, qa-easterns-routing-20260912-silver-spring-event-05; message_ids=f8b9a32a-1dd1-4570-a7e7-f180c20842b1, 71ac1e22-87a3-4fe0-bd9c-00dfc8da8fb8, 7062f4e9-43e3-42e3-9498-3619dcc8f118, 84549619-8da9-480c-8b88-67a5ad4c496b, d5944183-a9db-4c7d-aa6d-dd5e42639dd4

payload_hashes=ee7e5a6e7bb4252b0d28a7bfac8b874a6316c63b91e9889bebc1b3e251b4b24e, 93701b5e06177ed90037dde0c43055f35988f89b104e0000dd382324c558f7b6, 916f0ff1e4954b01f411096bfa7b0b724d85358a8a850b41ce86b32a119150fd, 1b7fa452096b0c31b1f5d3ac39a4dec9388a26cfc8621edb78fb63f3886bb00a, 800e4714bc6b714366567e0b0a763e1fc4b7d6ac419d52a8edb3cab82154cc7

snapshot={"assigned_dealer_id": "d1111111-1111-1111-1111-111111111111", "bank_account": "", "documents": "", "down_payment": "1000", "identification": "", "message_count": 5, "missing_qualification": ["identification", "proof_of_income", "bank_account"], "phone": "+12405553103", "purchase_timeline": "this week", "qualification_complete": true, "qualification_memory": "real_name: Buyer Cuatro; vehicle: Mustang; down payment: 1000; timeline: this week", "real_name": "Buyer Cuatro", "vehicle_type": "Mustang"}; location_snapshot={"city": "Silver Spring", "easterns_zone": null, "state": "MD", "zip_code": null}; dealer=Laurel; next_attempt_at=None
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.status, c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, l.id AS lead_id, l.canonical_phone, ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = \$1
ORDER BY ld.created_at NULLS LAST

qa-easterns-routing-20260912-washington

Mensajes: Me llamo Buyer Cinco | busco un Mustang | tengo 1000 de enganche | compro esta semana | Estoy en Washington, DC

conversation_id=qa-easterns-routing-20260912-washington-conversation; event_ids=qa-easterns-routing-20260912-washington-event-01, qa-easterns-routing-20260912-washington-event-02, qa-easterns-routing-20260912-washington-event-03, qa-easterns-routing-20260912-washington-event-04, qa-easterns-routing-20260912-washington-event-05; message_ids=8994448e-972e-49db-8428-5252caee399f, 49c1d1d7-1894-4b0d-8090-3eddf529d52, a7d7dcb3-2915-4c2a-b88e-8b375a3960bf, e77eef77-11f3-49f3-adcd-5544e7d6e069, c8fc757c-359f-4888-ba26-df66e3cc1f44

payload_hashes=e67f9035856bd700c14ba54117392422be021c231cfc3e3ed0dab4dc82deb486, bde4367b550a4edbebb1b776dd1ad109ad7d9651fe95dac2f015e37d11030d421, 78bcdbf4dbb49c201efcc129591528f314cb4bb84604b67ced387ce017a81e97, a4055afbebb16abc8e32a77522741d59836314b6d57e20604a7ef739797f68c7f, 3ef0056947e4811735bce7a3297f36f58f7e89e995b4dfcdfb5859658341aa17

snapshot={"assigned_dealer_id": "d1111111-1111-1111-1111-111111111111", "bank_account": "", "documents": "", "down_payment": "1000", "identification": "", "message_count": 5, "missing_qualification": ["identification", "proof_of_income", "bank_account"], "phone": "+12405553104", "purchase_timeline": "this week", "qualification_complete": true, "qualification_memory": "real_name: Buyer Cinco; vehicle: Mustang; down payment: 1000; timeline: this week", "real_name": "Buyer Cinco", "vehicle_type": "Mustang"}; location_snapshot={"city": "Washington", "easterns_zone": null, "state": "DC", "zip_code": null}; dealer=Sterling; next_attempt_at=None
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.status, c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, l.id AS lead_id, l.canonical_phone, ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = \$1
ORDER BY ld.created_at NULLS LAST

qa-easterns-routing-20260912-southern-md-1

Mensajes: Me llamo Buyer Seis | busco un Mustang | tengo 1000 de enganche | compro esta semana | Estoy en Waldorf, MD

conversation_id=qa-easterns-routing-20260912-southern-md-1-conversation; event_ids=qa-easterns-routing-20260912-southern-md-1-event-01, qa-easterns-routing-20260912-southern-md-1-event-02, qa-easterns-routing-20260912-southern-md-1-event-03, qa-easterns-routing-20260912-southern-md-1-event-04, qa-easterns-routing-20260912-southern-md-1-event-05; message_ids=4329fec4-5676-4be5-8a42-b80973c8b1f2, 10e0d533-d5dc-438e-9ff6-86c688f3d837, 9bdb34e1-6b57-4663-9a15-6578be00c720, e950d378-f9a1-44a2-a4fb-af2cdb3910cf, 381d59e8-20c5-4a25-8456-f227884e2347

payload_hashes=88b52f6b567254fad0fc61f64dcfc0a1594193650c63a46a90e449c16e96d394, 38d0baaea8ae302e4b174963200085781a723b763144adda5c8d6dbaea972fab, a22f482e445d512d19f755c0858dbd3af3e41f8c6836ebddf5abb1c6db56f8a97, 2371440474457a9e81f1bf28146899bdfbf18cffee117fd39120558a09d67e, 5fc044f77b8b51de43fc5f064cca963b8f0a4bb37e4bc0e4b997c931814d8196

snapshot={"assigned_dealer_id": "d1111111-1111-1111-1111-111111111111", "bank_account": "", "documents": "", "down_payment": "1000", "identification": "", "message_count": 5, "missing_qualification": ["identification", "proof_of_income", "bank_account"], "phone": "+12405553105", "purchase_timeline": "this week", "qualification_complete": true, "qualification_memory": "real_name: Buyer Seis; vehicle: Mustang; down payment: 1000;

timeline: this week", "real_name": "Buyer Seis", "vehicle_type": "Mustang"]; location_snapshot={"city": "Waldorf", "easterns_zone": null, "state": "MD", "zip_code": null}; dealer=Laurel; next_attempt_at=None
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.status, c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, l.id AS lead_id, l.canonical_phone, ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = \$1
ORDER BY ld.created_at NULLS LAST

qa-easterns-routing-20260912-southern-md-2

Mensajes: Me llamo Buyer Siete | busco un Mustang | tengo 1000 de enganche | compro esta semana | Estoy en La Plata, MD

conversation_id=qa-easterns-routing-20260912-southern-md-2-conversation; event_ids=qa-easterns-routing-20260912-southern-md-2-event-01, qa-easterns-routing-20260912-southern-md-2-event-02, qa-easterns-routing-20260912-southern-md-2-event-03, qa-easterns-routing-20260912-southern-md-2-event-04, qa-easterns-routing-20260912-southern-md-2-event-05; message_ids=5e15eb52-cc70-425d-a323-dc9a5ee52718, 38b633d8-73a9-4830-a36a-9e2a7d8a71bd, f1e8b91d-ab75-4666-baeb-4d1c3c3eae2, d2dee21f-4cec-475c-95e2-d4f2bb5202fc, b45a6cad-a86c-480c-a10f-dcdfe6ba2778

payload_hashes=5598ae6053cee89e9b7063d6ef5975c4e18da772c4a84f955f1efb1330d01efd, ec3777dc78d4867d6bdaa100c4ab081329ee1ef286c6f7c58f50409bb3823f56, 970a93d26f2900c2d2e1dca8ce91a622eb03c26e6f73f5d32ff42b0d74e39eac, d05e332c22e3b6af9d37a4724bc294a1dcfd986dedfd745c12650507dbd9ce1, fbbaf787621df1c3eb7b19cabbe40d2558f4aef07dd26c8a2fab3fb4b8b7529

snapshot={"assigned_dealer_id": "d1111111-1111-1111-1111-111111111111", "bank_account": "", "documents": "", "down_payment": "1000", "identification": "", "message_count": 5, "missing_qualification": ["identification", "proof_of_income", "bank_account"], "phone": "+12405553106", "purchase_timeline": "this week", "qualification_complete": true, "qualification_memory": "real_name: Buyer Siete; vehicle: Mustang; down payment: 1000; timeline: this week", "real_name": "Buyer Siete", "vehicle_type": "Mustang"]; location_snapshot={"city": "La Plata", "easterns_zone": null, "state": "MD", "zip_code": null}; dealer=STERLING; next_attempt_at=None
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.status, c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, l.id AS lead_id, l.canonical_phone, ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = \$1
ORDER BY ld.created_at NULLS LAST

qa-easterns-routing-20260912-virginia

Mensajes: Me llamo Buyer Ocho | busco un Mustang | tengo 1000 de enganche | compro esta semana | Estoy en Richmond, VA

conversation_id=qa-easterns-routing-20260912-virginia-conversation; event_ids=qa-easterns-routing-20260912-virginia-event-01, qa-easterns-routing-20260912-virginia-event-02, qa-easterns-routing-20260912-virginia-event-03, qa-easterns-routing-20260912-virginia-event-04, qa-easterns-routing-20260912-virginia-event-05; message_ids=9ca754df-a09d-4035-bfbb-fa602a057bd5, 7b395a8b-026c-468a-8083-733ceac847bc, eb952cc7-0002-4ded-8f9b-afd1569120ba, 1658eaaf-7aa5-4973-ba1b-1f7bd8dc4f28, c05044e9-023f-47fd-93c0-91fecdf7ccb8

payload_hashes=407eaaa52652d3c333e7932dd4d2aad00a92d0b320223e67d0cd34c2bfb4c993, 6e346fc7e1f87af75b5b5e5c4e2f10522133b9be8f9d1cbf8161bad5aa0c5137, cd7fe15e695f9e1aa78ace6406a6318e2b7c903be275c37ae6b5f91a2c6ce2f1, 4547360c2b44734b6c90e7679ae0a3a1d3a9c674b386b9ae431e22667bc9e92, cee7c0a52f11e2bdde582c3c208fb8e4541407883a7f018d55f43b41c45cae4a

snapshot={"assigned_dealer_id": "d1111111-1111-1111-1111-111111111111", "bank_account": "", "documents": "", "down_payment": "1000", "identification": "", "message_count": 5, "missing_qualification": ["identification", "proof_of_income", "bank_account"], "phone": "+12405553107", "purchase_timeline": "this week", "qualification_complete": true, "qualification_memory": "real_name: Buyer Ocho; vehicle: Mustang; down payment: 1000; timeline: this week", "real_name": "Buyer Ocho", "vehicle_type": "Mustang"]; location_snapshot={"city": "Richmond", "easterns_zone": null, "state": "VA", "zip_code": null}; dealer=STERLING; next_attempt_at=None
SELECT c.id AS conversation_db_id, c.ghl_conversation_id, c.status, c.qualification_snapshot, c.location_snapshot, c.next_attempt_at, l.id AS lead_id, l.canonical_phone, ld.dealer_id, ld.assigned_dealer_id, ld.status AS lead_dealer_status, ld.routing_reason
FROM conversations c
JOIN leads l ON l.id = c.lead_id
LEFT JOIN lead_dealers ld ON ld.lead_id = l.id
WHERE c.ghl_conversation_id = \$1
ORDER BY ld.created_at NULLS LAST

Normalizacion, correcciones y regresiones

Incidentes de normalizacion

Campo	Resultado
real_name	Solo se guardo con declaracion explicita; texto del bot o ubicaciones no se usaron como nombre.
vehicle_type	Mustang y truck se interpretaron con evidencia; Mustang aislado contaminante no creo lead.
down_payment	No se infirio; Easterns uso 1000 solo como enganche declarado.
telefono	Stafford uso el telefono preexistente y canonico.
documents / identification / bank_account	Quedaron vacios/null sin evidencia explicita; no se mezclaron campos.

Correcciones aplicadas despues de capturar evidencia

Hallazgo	Causa raiz	Correccion	Reprueba
Easterns dependia de ciudades codificadas	Listas en el servicio.	Migracion agrega easterns_routing_zone a locations y runtime consulta BD.	Easterns 8/8; unitarias 72/72
Firma del process-due del harness	Endpoint sin body requiere header compartido.	Correccion solo en harness; no era defecto de negocio.	Stafford 7/7 + Easterns 8/8
Frontend sin API_URL en primer arranque	Build local incompleto.	Rebuild con API_URL local.	Login/dashboard PASS
Fixtures contaminaban nombre con ciudad	Normalizador rechazo ubicaciones como nombres.	Nombres neutrales declarados.	Easterns 8/8

Regresiones ejecutadas

Suite completa: 27 archivos PASS; 328 tests PASS y 4 omitidos en modo normal. Suite de resiliencia con RUN_RESILIENCE_STRESS: 4/4 PASS, incluyendo 150 webhooks concurrentes, deduplicacion, orden actualizado-antes-que-base y rollback por conexion fallida. Stafford se repitio despues del rebuild de API y el frontend se verifico despues del rebuild.

Action Pre Owned Cars

Codigo	Nombre	Grupo	Idioma
ACTION-CARS-EN	Action Pre Owned Cars English	Action Pre Owned Cars	en
ACTION-CARS-ES	Action Pre Owned Cars Español	Action Pre Owned Cars	es

Frontend local: {"login_status": "PASS", "dashboard_status": "PASS", "url": "http://127.0.0.1:3000/app", "selected_dealer": "Offlease Motors Stafford", "qa_phone_visible": true, "qa_vehicle_visible": true, "screenshot": "tmp/qa-whatsapp-stafford-20260912/frontend-stafford-dashboard.png"}

dealerADMIN

ReportesWebhook activoESENCerrar sesión

D

DEALER ACTIVO

Offlease MotorsStafford

DEALERS

Action P...1Copiar

Action P...1Copiar

Arlington...1Copiar

Easterns...1Copiar

Easterns...4Copiar

Easterns...2Copiar

Easterns...2Copiar

Easterns...4Copiar

Easterns...5Copiar

Koons A...1Copiar

Koons A...1Copiar

Koons d...1Copiar

Offleas...14Copiar

Offlease...1Copiar

☒ Offlease...4Copiar

OPERACIONES / COLA

Cola de trabajo de leads

PendientesEnviados

Revisa, prepara y despacha leads calificados.

Offlease Motors Stafford4 leads pendientes

Flujo de webhook0 seleccionados

+ Agregar lead manual

Carga masiva

LEAD	VEHICULO	CALIFICACIÓN	RECIBIDO	ACCIONES
☐ Pedro Infante +13019876543 Pedro Infante +13019876543, Troca, ID: LIC-9900.	Troca Offlease Motors Stafford	down no indicado ID LIC-9900 cuenta bancaria no indicada documentos no indicados tiempo no indicado	LLEGÓ A DEALERADMIN 21 de ago, 4:21 p. m.	Copiar a... Marcar enviado Eliminar Copiar
☐ Sofia Qa +12405550120 Sofia Qa +12405550120 Mustang, 1000 down, proof of income: yes, wants to buy this month.	Mustang Offlease Motors Stafford	1000 ID no indicado cuenta bancaria no indicada prueba de ingresos this month	LLEGÓ A DEALERADMIN 12 de sept, 9:06 a. m.	Copiar a... Marcar enviado Eliminar Copiar
☐ Whatsapp Contact +12405552001 Lead +12405552001 Mustang.	Mustang Offlease Motors Stafford	down no indicado ID no indicado cuenta bancaria no indicada documentos no indicados tiempo no indicado	LLEGÓ A DEALERADMIN 12 de sept, 12:35 p. m.	Copiar a... Marcar enviado Eliminar Copiar
☐ Whatsapp Contact +12405552002 Lead +12405552002 truck.	truck Offlease Motors Stafford	down no indicado ID no indicado cuenta bancaria no indicada	LLEGÓ A DEALERADMIN 12 de sept, 12:35 p. m.	Copiar a... Marcar enviado Eliminar Copiar

Captura tomada del dashboard local Stafford con telefono y Mustang visibles.

Conclusiones y artefactos

Todos los criterios de aceptacion quedaron PASS. No quedan fallos abiertos ni bloqueos de infraestructura. La evidencia conserva IDs, hashes, snapshots, estados, ventanas simuladas y consultas SQL; no contiene secretos HMAC ni credenciales.

Artefactos publicados

Artefacto	Contenido
tmp/qa-whatsapp-stafford-20260912/evidence.json	Stafford: payload hashes, IDs, snapshots, estados, ventanas, SQL y resultados.
tmp/qa-easterns-routing-20260912/evidence.json	Easterns: ocho conversaciones, locations, dealer, routing_reason, IDs, hashes y SQL.
tmp/qa-whatsapp-stafford-20260912/frontend-stafford-dashboard.png	Captura del dashboard local.
apps/api/src/database/migrations/1710000021000-EasternsLocationRoutingRules.ts	Clasificacion geografica persistida en locations.

Los datos de prueba son locales y deterministas. No se borro informacion fuera de los namespaces QA.